Material Characterization and Instrumentation Techniques

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Outline

1. Introduction
2. Universal testing machine (UTM)
3. Hardness determination
   - Rockwell Hardness Testing
   - Brinell Hardness Testing
   - Vickers Hardness Testing
4. Magnetic Particle Testing
5. Ellipsometry
6. Current-Voltage (IV) characterization

Material Characterization:

- Mechanical Characterisation
  - Tensile and yield strength
  - Ductility
  - Toughness
  - Hardness
- Optical Characterisation
- Current-Voltage (IV) characterisation
- Surface characterisation
  - Roughness & Crystallinity
  - Particle distribution
  - Magnetic properties

Instrumentation Techniques:

- X-ray Diffractometer
- Crystallite size determination by Scherrer equation
- Atomic Force Microscopy (AFM)
- X-ray photoelectron spectroscopy (XPS)
- Scanning Electron Microscopy (SEM)
- Transmission Electron Microscopy (TEM)

Traditionally materials are classified into:

- Metals: (steel, copper, aluminum)
- Polymers: (cloth, plastic)
- Ceramics: (pottery, porcelin)
- Composites: (concrete, plywood)
- Semiconductors: (Si, Ge)

Macroscopic characterization technique:

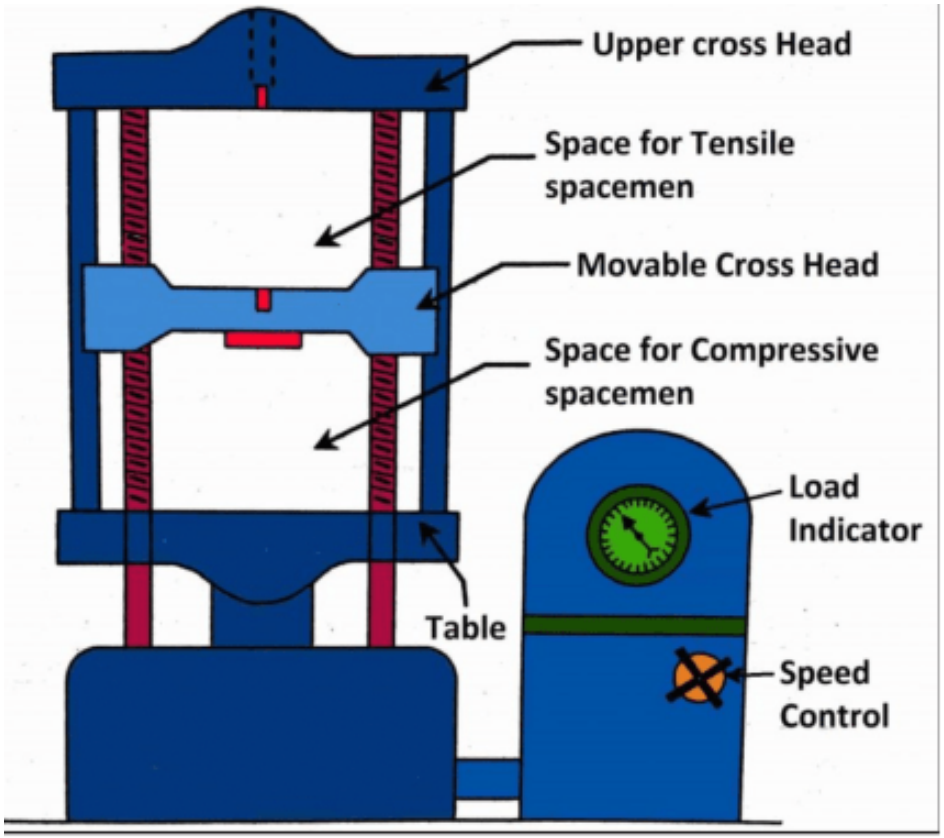
Tensile and yield strength, ductility, toughness and hardness that can be analyzed using specific instrument like Universal Testing Machine (UTM).

Microscopic characterization technique:

Surface morphology, compositions, grain size, microscopic structure, optical properties etc, can be analyzed using technical instruments like Scanning Electron Microscope (SEM), Transmission Electron Microscope (TEM), Atomic Force Microscope (AF), X-ray Diffractometer (XRD) etc., to understand the materials.

Universal testing machine (UTM)

A Universal testing machine (UTM) is used to test the mechanical properties (tension, compression etc.) of a given test specimen by exerting tensile, compressive or transverse stresses.





Construction of UTM

- The arrangement of the test specimen and the exertion of the load are held in the *loading unit*.
- The variations in the application of the load and the corresponding test result are obtained from the *control unit*.

Loading Unit:

- Load frame: Consists of a table (where the specimen is placed for the compression test)
- Upper crosshead: used to clamp one end of the test specimen
- Lower crosshead: is the movable crosshead whose screws can be loosened for height adjustment and tightened.
- Elongation scale: The relative movement of the lower and upper table is measured by an elongation scale which is provided along with the loading unit.

Control Unit:

- *Hydraulic power unit:* consists of an oil pump that provides non-pulsating oil flow into the main cylinder of the load unit.
- This flow helps in the smooth application of load on the specimen.
- The oil pump in a hydraulic power unit is run by an electric motor.
- *Load measuring unit:* has a pendulum dynamometer unit that has a small cylinder with a piston which moves with the non-pulsating oil flow.
- The pendulum is connected to the piston by pivot lever. The pivot lever deflects based on the load applied to the specimen.

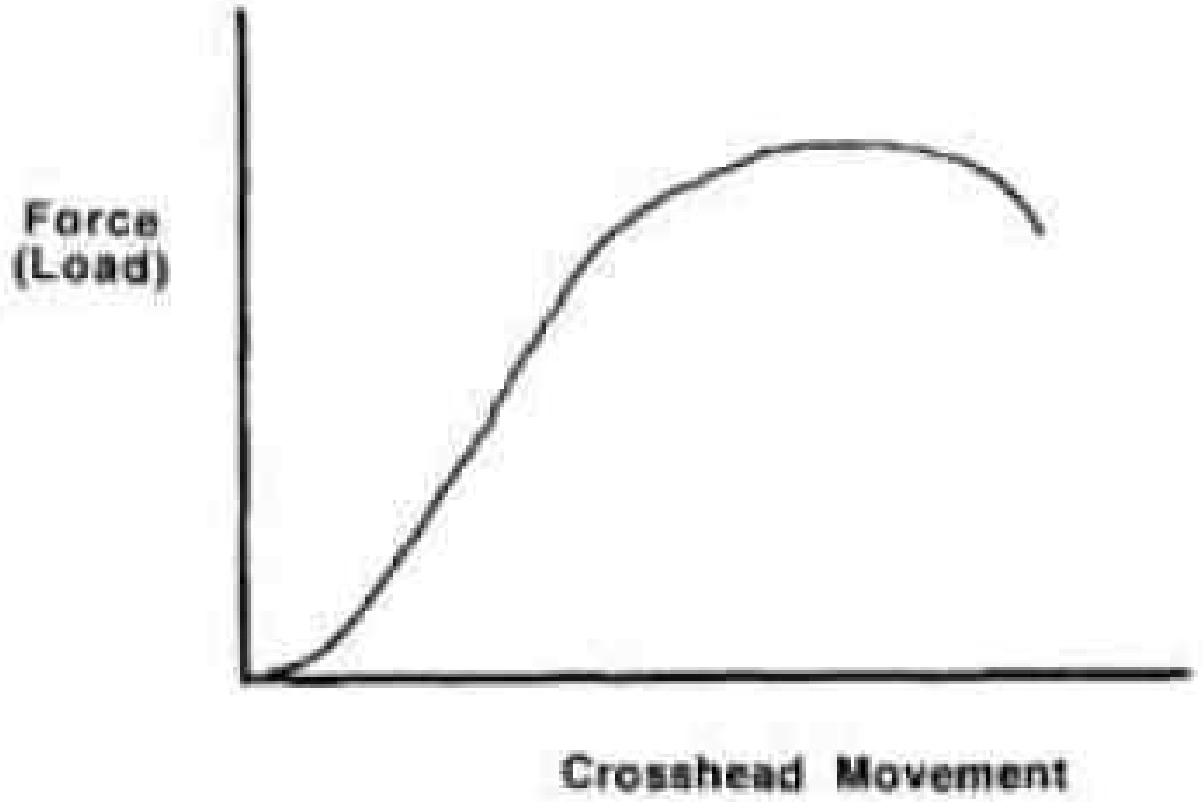
- This deflection is converted to the load pointer and displays as the load on the dial.
- The range of load application can be adjusted by means of a knob in the load measuring unit (0-100 kN; 0-250 kN; 0-500 kN and 0-1000 kN).
- *Control devices:* can be electric or hydraulic.
- Electric control devices make use of switches to move the crossheads and switch on/off the unit.
- A hydraulic control device consists of two valves, right control valve and left control valve or release valve.
- A right control valve is used to apply load on the specimen. The left control valve is used to release the load application.

Theory of UTM

- Tension test is widely used to provide basic design information on the strength of the materials.
- In the tension test a specimen is subjected to a continually increasing uniaxial tensile force while simultaneous observations are made of the elongation of the specimen.
- A stress-strain curve is plotted from the load-elongation measurements.
- The parameters which are used to describe the stress-strain curve of a material are the tensile strength, yield point, percent elongation and reduction of area.
- The first two gives strength parameters and the last two indicate the ductility of the material.

Working of UTM

- The standard tests performed by UTM are: Tensile test, compression test, adhesion tests, pull-out tests, bending test, hysteresis test etc.
- UTM provides the value of load application and the respective displacements.
- From the observed value, the load deflection graph is obtained.
- With the load value in the Y-axis and the displacement in the X-axis graph will be plotted.
- The displacement is actually the movement of the crossheads during the load application.
- From the load deflection graph, the stress-strain analysis, modulus of elasticity, yield strength of the specimen tested can be determined.


Force
(Load)
Crosshead Movement

Analysis of UTM data

- The UTM used for tensile test consists of a hydraulic device for applying a steadily increasing load to a test piece of metal. The load is indicated on a calibrated dial. The material specimen is firmly held in the grips of the testing machine without slippage when the load is applied.
- The specimen is marked with a 'gauge-length', shorter than the distance between the grips and this will be considered as L_0. This L_0 should be measured every time as and when the reading of the load is taken. The normal stress (load applied) - strain (displacement) curve can be plotted and defined as follows.

$$\text{Stress } T = \frac{\text{Load at any instant in newtons}}{\text{Original cross-sectional area in mm}^2} = \frac{P}{A_0} \quad (\text{in N/mm}^2)$$

$$\text{Strain } \varepsilon = \frac{\text{Increase of gauge length}}{\text{Original gauge length}} = \frac{L - L_0}{L_0}$$

Ductile material is tested under tension:

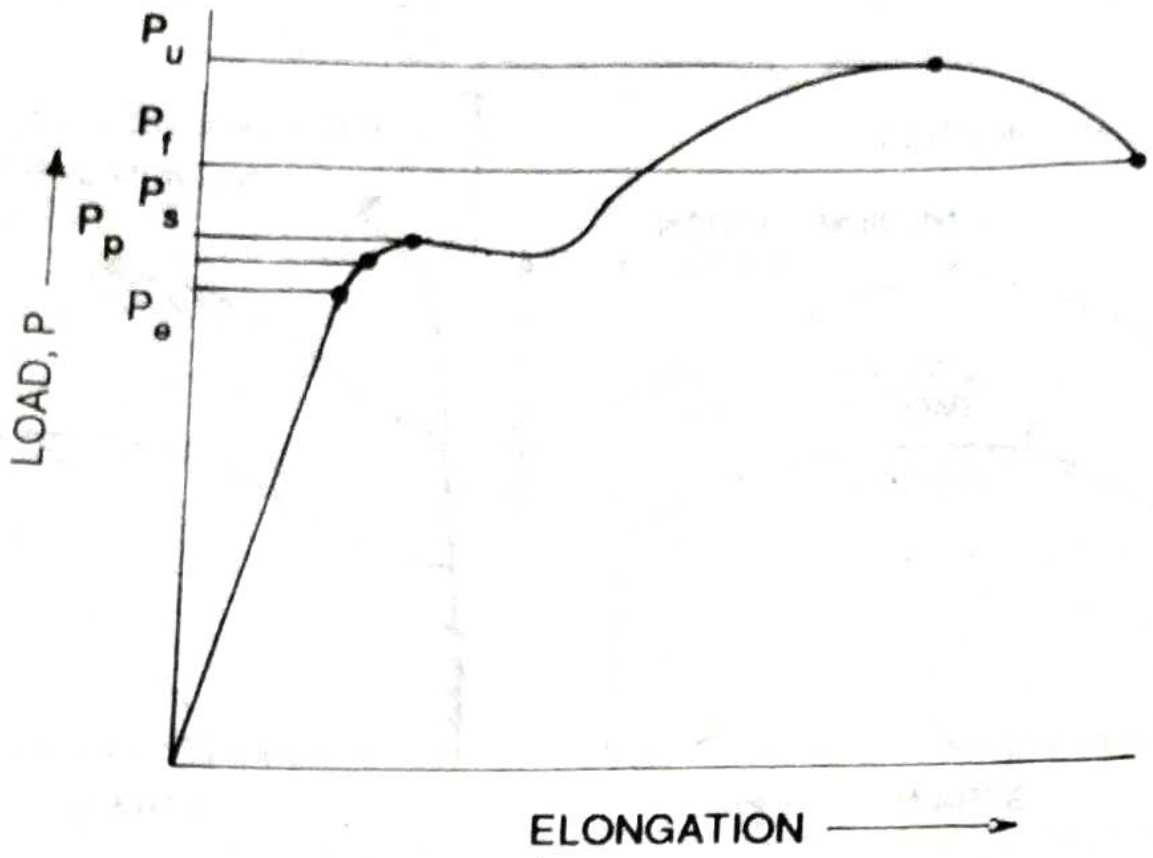


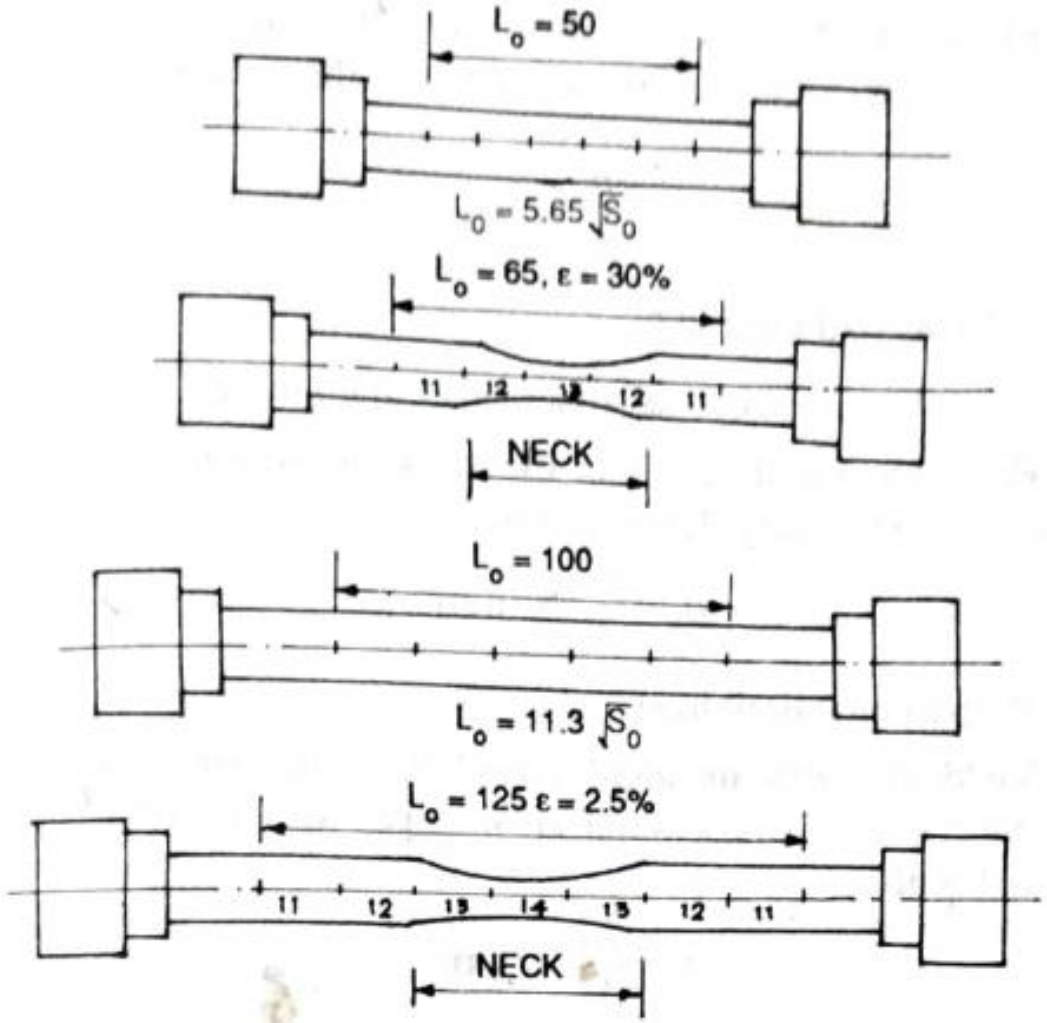
Figure: Stress – strain graph for a ductile material.

Ultimate tensile strength:

$$T_u = \frac{\text{Ultimate Load}}{\text{Original area of cross-section}} = \frac{P_u}{A_0} \qquad (\text{N/mm}^2)$$

where P_u = maximum load where necking down occurs and A_0 = initial cross sectional area of specimen in mm^2.

Tensile test for short and long test rods:



It can be observed from the figure that the length of a neck portion will be about equal for long and short test rods but the neck elongation will be higher percentage of the ultimate strain in short rod (30%) and a smaller percentage in long rod (2.5%). It means a higher value for ultimate strain in short test rod than in long test rod.

Other mechanical parameters can be analyzed from stress-strain graph:

- Elastic limit: It is the maximum stress that the metal can withstand without experiencing a permanent strain when the load is removed. It is given by

$$T_e = \frac{P_e}{A_0} \qquad (\text{N/mm}^2)$$

where P_e = elastic load limit in N and A_0 = initial cross sectional area of specimen in mm^2.

- Proportional limit: It is the stress at which the stress strain curve deviates from linearity and is given by

$$T_p = \frac{P_p}{A_0} \qquad (\text{N/mm}^2)$$

where P_p = Proportional limit load.

- Yield point: It is the stress at which the material elongates or yields without an increase in load. It is obtained at yield point and is given by

$$T_s = \frac{P_s}{A_0} \qquad (\text{N/mm}^2)$$

where P_s = load at the yield point.

- Reduction of area: It is the ratio of the decrease in the cross- sectional area of the test piece after fracture of the original area, expressed in percent.

$$\text{Reduction of area (in \%)} = \frac{d_0^2 - d^2}{d_0^2} \times 100$$

Where d_0 = original cross sectional diameter, d = minimum diameter of the necked portion.

- Ductility: Ductility of a material cannot be expressed exactly from tensile test. However, valuable information is obtained from the test curve and by the percentage elongation (i.e., strain) and percentage reduction in area of the test piece.
- Stiffness: The modulus of elasticity is a measure of stiffness of the material and is given by

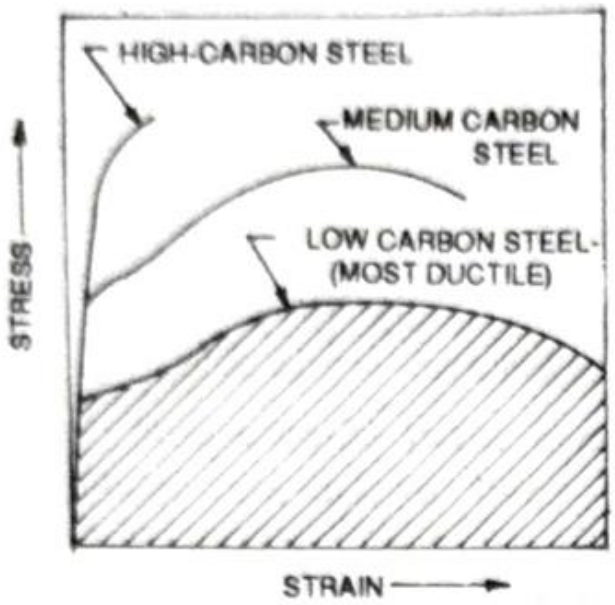
$$E = \frac{T}{\epsilon} = \frac{P_p L}{\epsilon A_0}$$

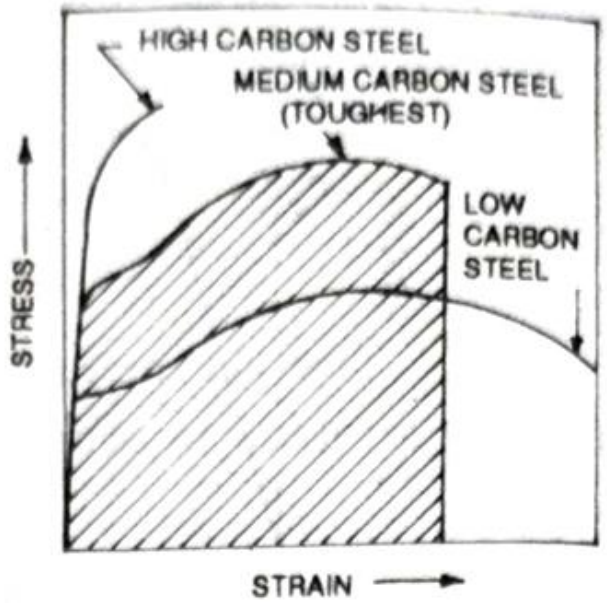
where P_p = load at proportional limit, L = gauge length.

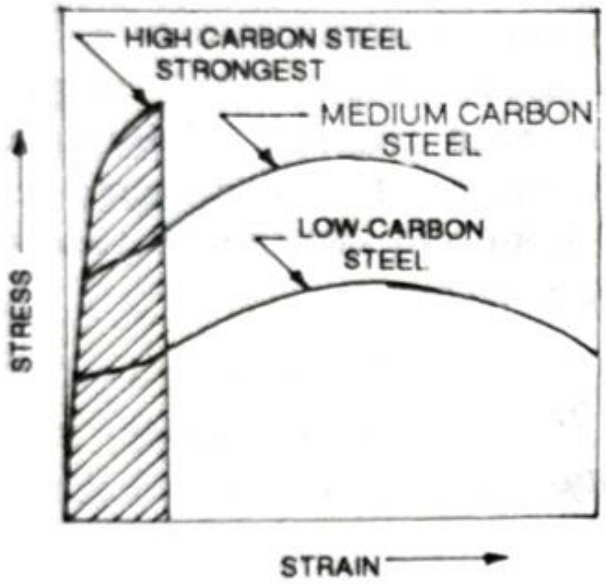
- Toughness: Toughness is a combination of strength and plasticity. A tough material can take hard blows without rupturing. Toughness is often defined as a material's ability to absorb energy without cracking. The modulus of toughness is the amount of energy required to cause failure in unit volume of a material and this is represented by the total area under the stress- strain curve. (see figure below)
- Rupture: Rupture is expressed by the breaking strength of the material of the test piece in the tensile test.

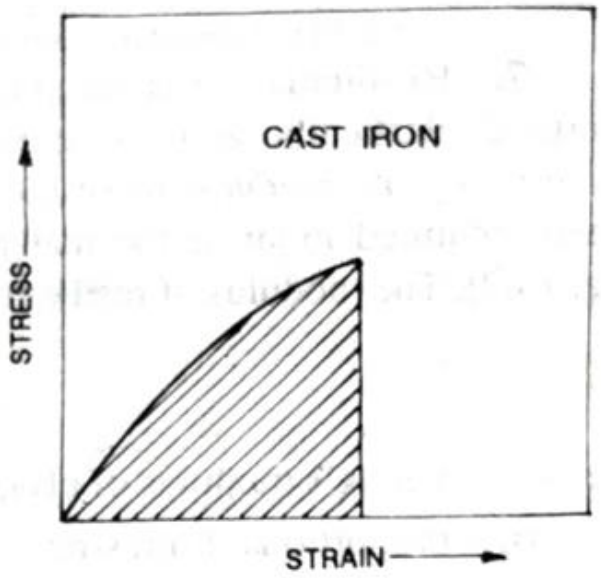
$$= \frac{P_f}{A_0}$$

where P_f = load at failure.


HIGH-CARBON STEEL
MEDIUM CARBON STEEL
LOW CARBON STEEL-(MOST DUCTILE)
STRESS
STRAIN


HIGH CARBON STEEL
MEDIUM CARBON STEEL (TOUGHEST)
LOW CARBON STEEL
STRESS
STRAIN


HIGH CARBON STEEL STRONGEST
MEDIUM CARBON STEEL
LOW-CARBON STEEL
STRESS
STRAIN


CAST IRON
STRESS
STRAIN

Hardness determination

Hardness is a characteristic of a material, not a fundamental physical property.

> *It is defined as the resistance to indentation, and it is determined by measuring the permanent depth of the indentation.*

When we apply a fixed force with a given indenter, the smaller the indentation, indicate that the material is harder. Indentation hardness value is obtained by measuring the depth or the area of the indentation.

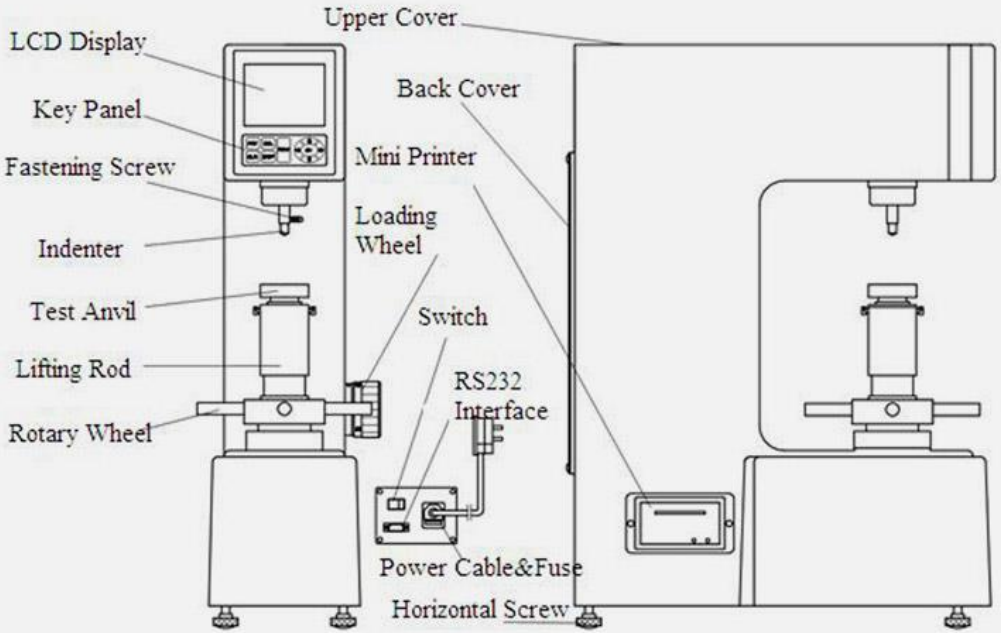
- Materials such as hardened bearing steels have small grain size and can be measured using the **Rockwell scale** due to the use of diamond indenters and high PSI loading.
- Materials such as cast iron and powder metals will need a much larger indenter such as used with **Brinell scales**.
- Very small parts or small sections may need to be measured on a micro hardness tester using the **Vickers or Knoop Scale**.

1. Rockwell Hardness Testing

The Rockwell method measures the permanent depth of indentation produced by a force/load on an indenter. First, a preliminary test force (commonly referred to as preload or minor load) is applied to a sample using a diamond or ball indenter. This preload breaks through the surface to reduce the effects of surface finish.

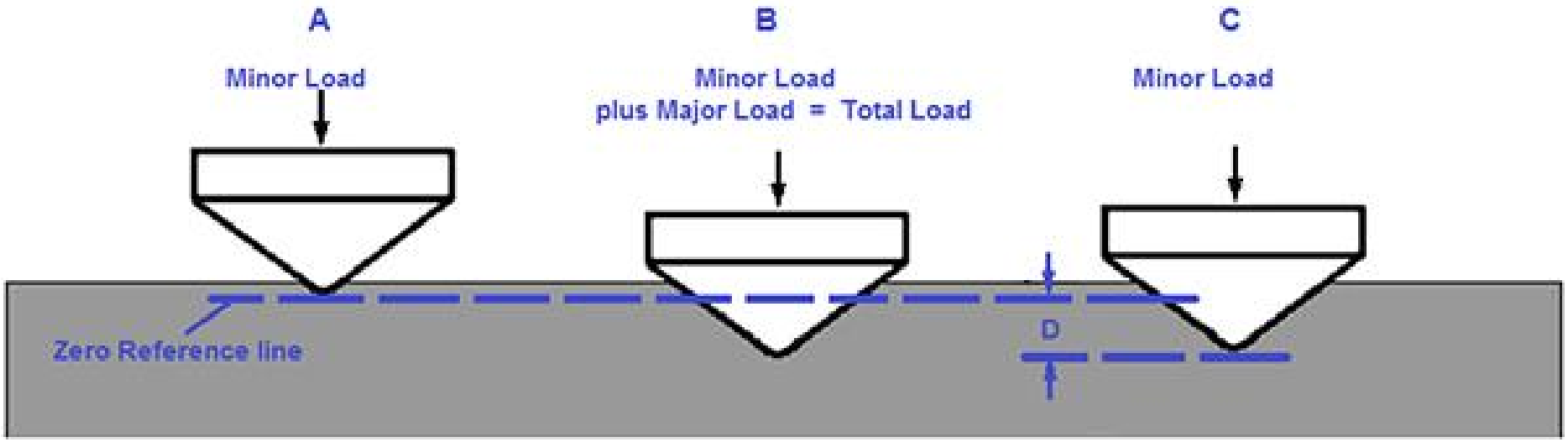






After the preload, an additional load, call the major load, is added to reach the total required test load. This force is held for a predetermined amount of time (dwell time) to allow for elastic recovery.

Rockwell hardness value is derived from the difference in the baseline and final depth measurements. This distance is converted to a hardness number.



Preliminary test loads (preloads) range from 3 kgf (used in the "Superficial" Rockwell scale) to 10 kgf (used in the "Regular" Rockwell scale). Total test forces range from 15kgf to 150 kgf (superficial and regular) to 500 to 3000 kgf (macrohardness).

Test Method Illustration:

- A = Depth reached by indenter after application of preload (minor load)
- B = Position of indenter during Total load, Minor plus Major loads
- C = Final position reached by indenter after elastic recovery of sample material
- D = Distance measurement taken representing difference between preload and major load position. This distance is used to calculate the Rockwell Hardness Number.

2. Brinell Hardness Testing

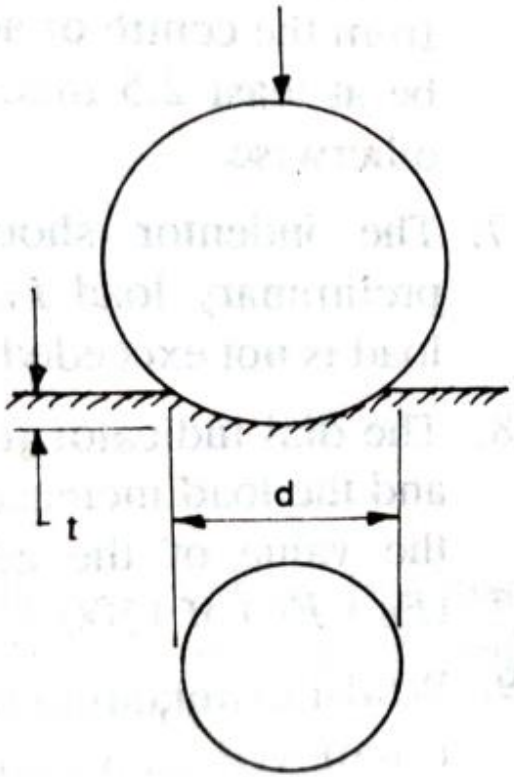
- The Brinell hardness test method is used to test materials that have a structure that is too coarse or that have a surface that is too rough to be tested using another test method, e.g., castings and forgings.
- Brinell testing often use a very high test load (3000 kgf) and a 10mm diameter indenter so that the resulting indentation averages out most surface and sub-surface inconsistencies.

The method:

- The Brinell method applies a predetermined test load (P) to a carbide ball of fixed diameter (D) which is held for a predetermined time period and then removed.
- The resulting impression is measured with a specially designed Brinell microscope or optical system across at least two diameters – usually at right angles to each other and these results are averaged (d).
- The Brinell hardness number is obtained by dividing the test load P by curved surface area of the indentation.
- This curved surface is assumed to be portion of the sphere of diameter D.
- The Brinell Hardness Number is given by

$$\text{BHN} = \frac{P}{A}$$

Where P = load in N, A = area of contact between the ball and the indentation in mm^2.



If the deformation of the ball during the indentation is neglected, the surface contact is a spherical segment whose area is given by

$$A = \pi D t$$

Where, D = diameter of the ball, t = depth of the impression.

If d is the diameter of the impression then by geometry,

$$t = \frac{D - \sqrt{D^2 - d^2}}{2}$$

$$\therefore \text{ BHN} = \frac{P}{A} = \frac{2P}{\pi D(D - \sqrt{D^2 - d^2})}$$

Disadvantages of Brinell test:

- It produces a fairly large indentation, particular on soft materials.
- It cannot be used near the edge of the specimen.
- It cannot be used on thin metals since reaction from the test anvil might influence test results.

However, irrespective of these limitations, it does give a linear scale of hardness and is particularly useful for design work.

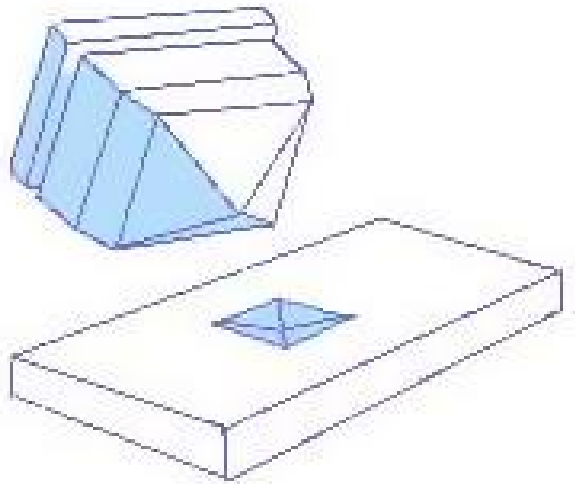
Common test forces range from 500kgf often used for non-ferrous materials to 3000kgf usually used for steels and cast iron.

There are other Brinell scales with load as low as 1kgf and 1mm diameter indenters but these are infrequently used.

3. Vickers Hardness Testing

- The Vickers hardness test method, also referred to as a micro hardness test method, is mostly used for small parts, thin sections.
- The Vickers method is based on an optical measurement system.
- The Microhardness test specifies a range of light loads using a diamond indenter to make an indentation which is measured and converted to a hardness value.
- It is very useful for testing on a wide type of materials, but test samples must be highly polished to enable measuring the size of the impressions.
- A square base pyramid shaped diamond is used for testing in the Vickers scale.
- Typically loads are very light, ranging from 10gm to 1kgf, although Macro Vickers loads can range up to 30 kg or more.
- The Microhardness methods are used to test on metals, ceramics, composites - almost any type of material.

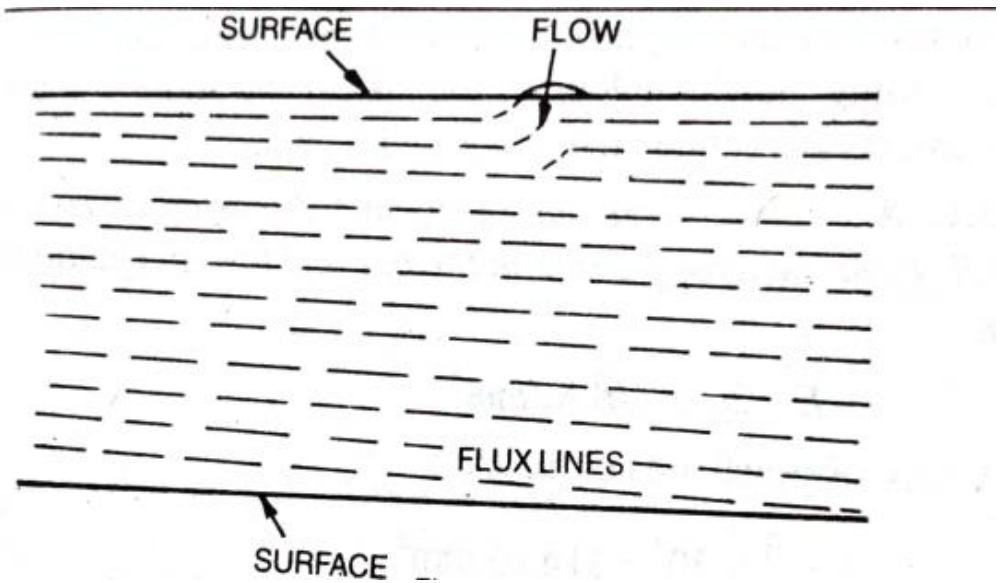
- The sample preparation should make the specimen's surface smooth to permit a regular indentation shape and good measurement, and to ensure the sample can be held perpendicular to the indenter.
- The prepared samples are mounted in a plastic medium to facilitate the testing.
- The indentations should be as large as possible to maximize the measurement resolution. (Error is magnified as indentation sizes decrease).



Magnetic Particle Testing

- Magnetic particle inspection is based on the magnetic lines of flux or force lines that can be generated within the test specimen.
- These force lines are parallel to each other while travelling through the test specimen if there is no surface or sub-surface defects.
- Force lines go through the material without any deviation. However, if there is a surface defect, the magnetic lines of force or the flux lines, will create a small north-south pole area at the location of the surface defect causing the iron powder that is used in magnetic particle inspection to collect around the surface defect.
- This surface defects is called a magnetic leakage or an indication.
- In this technique magnetic field can be produced by strong magnet or an electromagnet (using DC power).

- Magnetic particle inspection can also indicate sub-surface defect if these defects are close to the surface.
- These sub-surface defects disrupt the magnetic lines of force and bend them around the defect.
- A sub-surface defect is indicated by the magnetic powder collecting around the broken lines of force at the surface of the test specimen. The subsurface defect is shown in the below figure.



- Two types of surface defects can be detected by magnetic particle inspection, defects that are **parallel** to the interface of the work piece and defects that are **perpendicular** to the interface or sides of the test specimen.
- Magnetic lines of force must cross the defect in a perpendicular line in order to derive the benefit from magnetic particle inspection.
- If the defect is parallel to the lines of force, there is a possibility that the indication of a surface or a sub surface flow in the test specimen would not be noticeable.
- In order to check a test specimen fully, both longitudinal and perpendicular indications must be checked.
- The perpendicular indications are inspected by using a circular coil which are travelling through the specimen longitudinally.
- Any defect that is perpendicular to these magnetic lines of flux is shown in below figure (perpendicular defect).
- The power requirements for electromagnets are low voltage and high ampere DC.

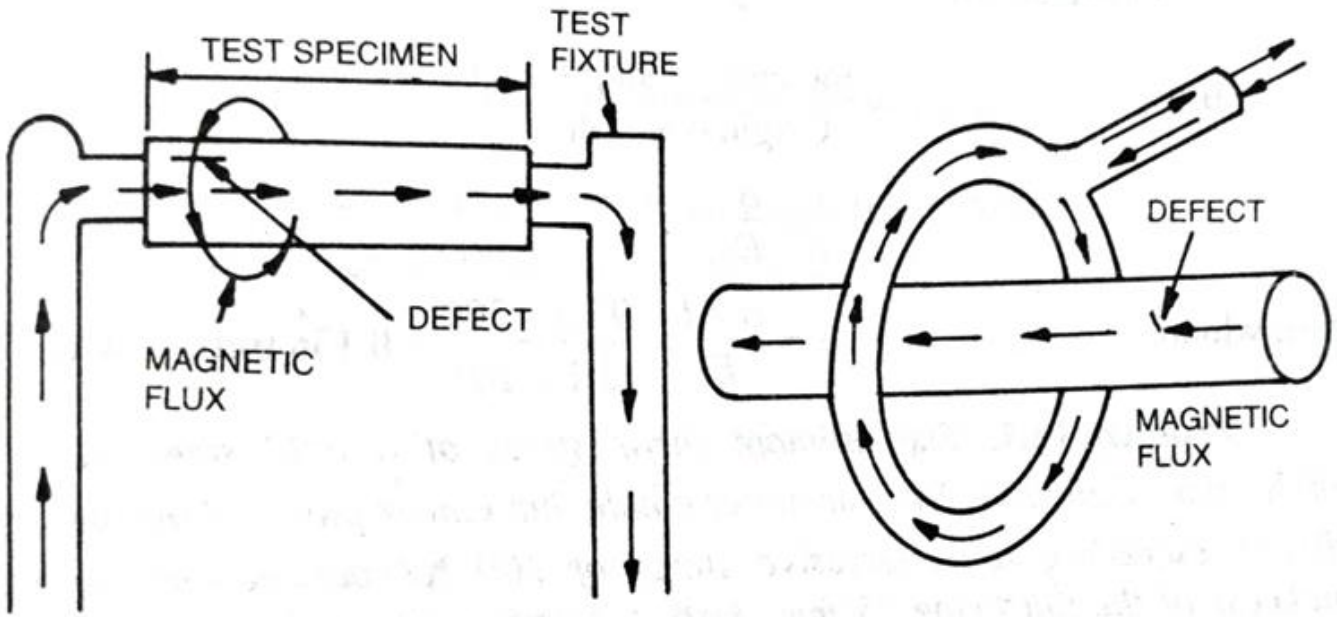


Figure: Longitudinal and perpendicular defects.

Ellipsometry

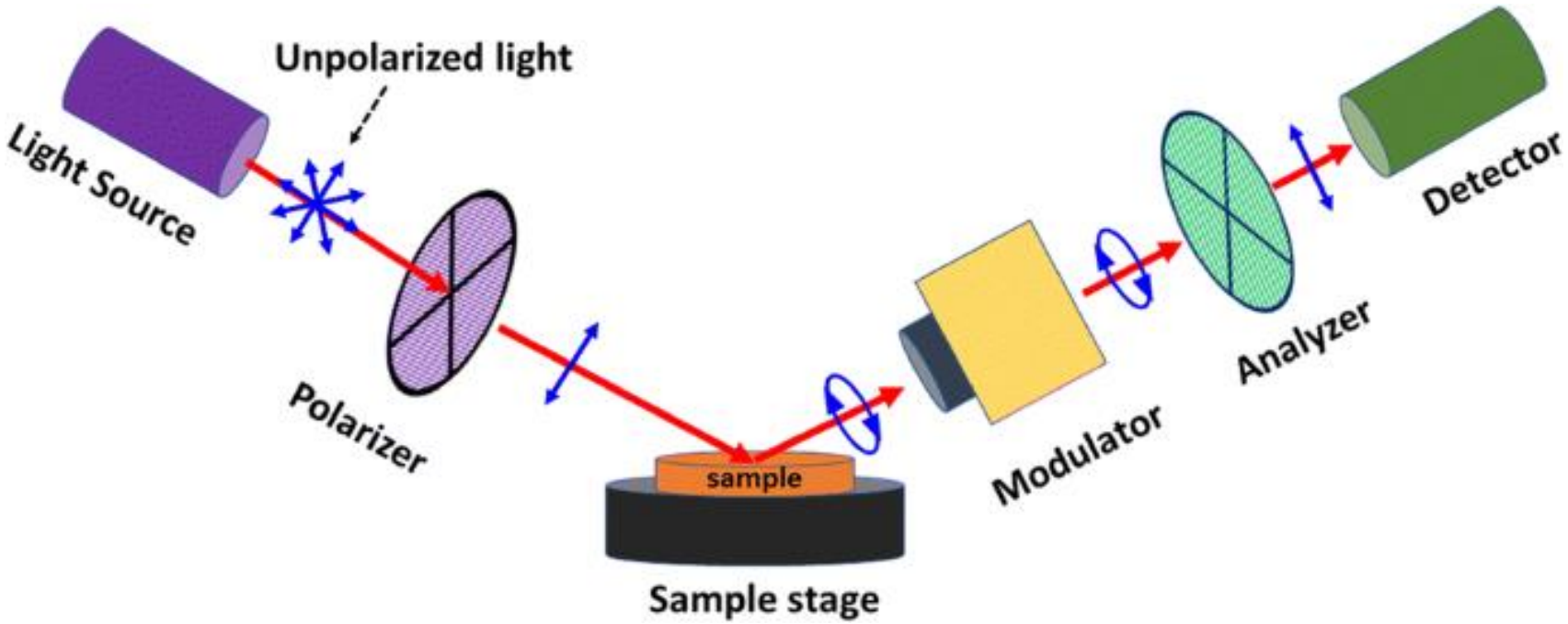
(for optical and dielectric characterization)

- Ellipsometry is an optical technique for investigating the optical or dielectric properties (complex refractive index or dielectric function and extinction coefficient) and thickness of thin films.
- The material specimen should be in the form of thin film with the thickness of the order of few mm.
- Ellipsometry measures the change of polarization upon reflection or transmission and compares it to a model.
- It is an optical technique used to determine when a parallel beam of monochromatic light of known polarization is incident on a semi transparent thin film.
- The amplitude of the two perpendicular components of the polarized light and the relative phase difference between them change due to reflection.
- Optical properties cause changes in the ellipticity of the polarized light and hence the name *ellipsometry*.

Construction:

A typical ellipsometer consists of the following parts:

- *Light source:* It can be either a laser or a lamp (Xenon lamp) emitting continuous spectrum with a monochromator and a focusing system.
- *Polarizer:* It consists of a quartz window.
- *Modulator:* It modulates the light beam polarization at 50kHz with the help of computer and control electronics.
- *The analyzer:* It is a linear polarizer which resolves the polarization state of the reflected beam.
- *The detector:* The detector can be any device able to measure the intensity of the light wave. Generally, two types of detectors are employed viz. photomultipliers and InGaAs photodiodes for UV-visible and NIR applications respectively.


Unpolarized light
Light Source
Polarizer
sample
Sample stage
Modulator
Analyzer
Detector

Working:

- A light from the source is linearly polarized using a polarizer and is made to be incident on the surface of the sample.
- The linearly polarized light after reflection from the sample surface becomes elliptically polarized, and travels through a modulator.
- The function of modulator is to introduce a difference in the phase velocity between the ordinary and extraordinary rays leading to a modulated phase shift.
- Analyzer examines the incoming lights having different phase velocities and sends the light to the detector.
- The detector converts the light into respective electronic signal to determine the polarization of the reflected light.
- This information is compared with the known input polarization caused by the reflection at the sample.

Calculation:

- For the measurement of complex reflectance ratio ρ of a system, which may be parametrized by the amplitude component Ψ and the phase difference Δ.
- The polarization state of the light incident upon the sample may be decomposed into an s and a p component.
- The s component is oscillating perpendicular to the plane of incidence and parallel to the sample surface, and the p component is oscillating parallel to the plane of incidence.
- The amplitudes of the s and p components, after reflection and normalized to their initial value, are denoted by R_s and R_p respectively.
- The angle of incidence is chosen close to the Brewster angle of the sample to ensure a maximal difference in R_p and R_s.

- Ellipsometry measures the complex reflectance ratio ρ (a complex quantity), which is the ratio of R_p over R_s:

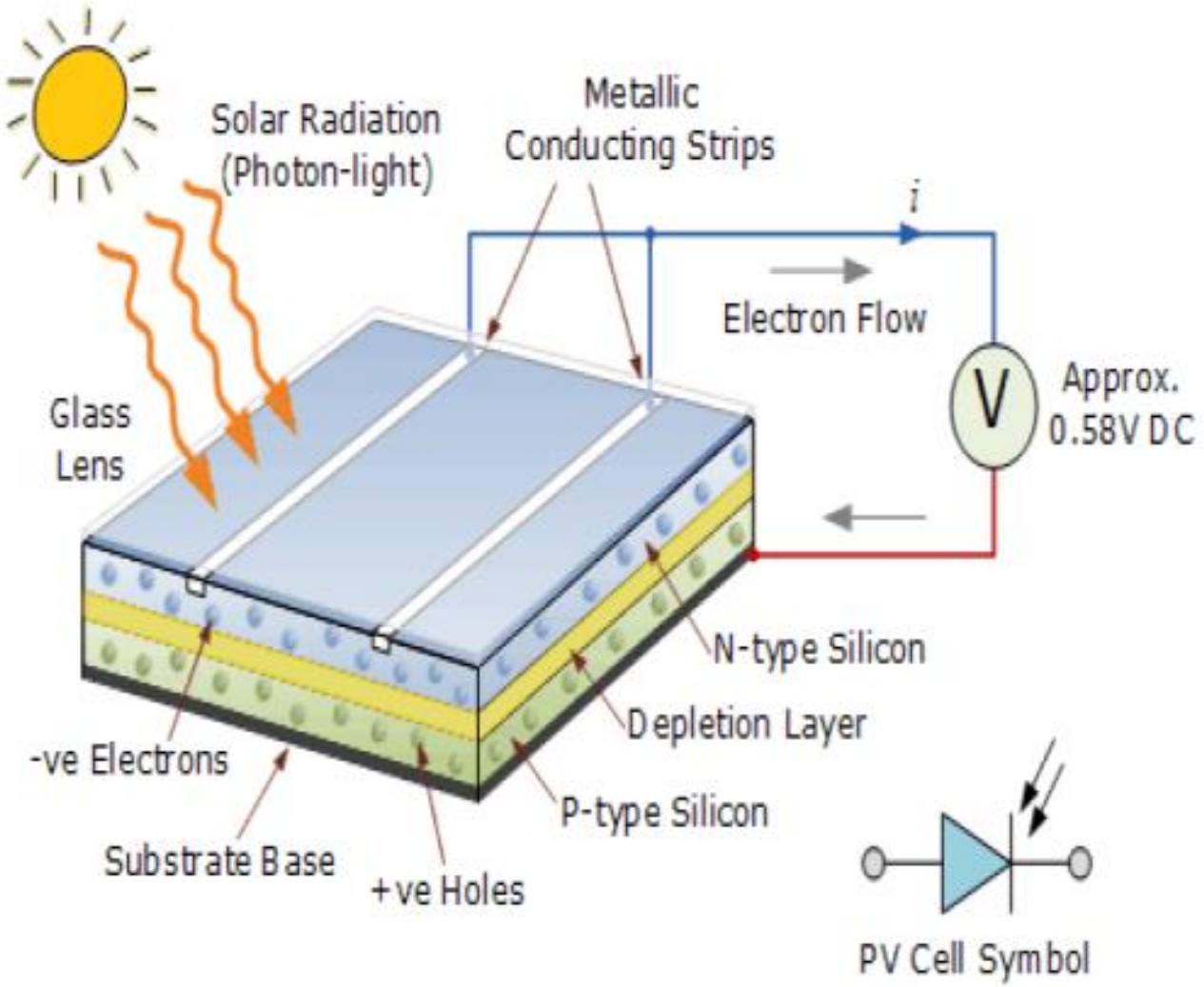
$$\rho = \frac{R_p}{R_s} = \tan\Psi\, e^{i\Delta}$$

  where R_S and R_P are the amplitudes of the s and p components of the polarized light, $\tan\Psi$ = modulus of amplitude reflection ratio and Δ = phase difference between s and p polarized reflected light.
- From the ellipsometric parameters obtained from the experiment one can determine the refractive index and the thickness of the sample.

Current-Voltage (IV) characterization

- The conducting property (conductivity) of a Semiconducting material can be analysed using current-voltage (I-V) characterization technique.
- Let us consider a solar cell of photovoltaic (PV) device made up of semiconductor photo diode for I-V analysis.
- Current-Voltage (IV) characterization of a semiconducting material is the major technique to study the performance of solar efficiency of a solar cell.
- The solar Cell I-V characteristic curve shows the current and voltage (I-V) characteristics of a particular photovoltaic (PV) cell, module or array.
- It gives a detailed description of its solar energy conversion ability and efficiency.
- The output performance and efficiency of the device (solar cell) is analyzed using I-V curve.

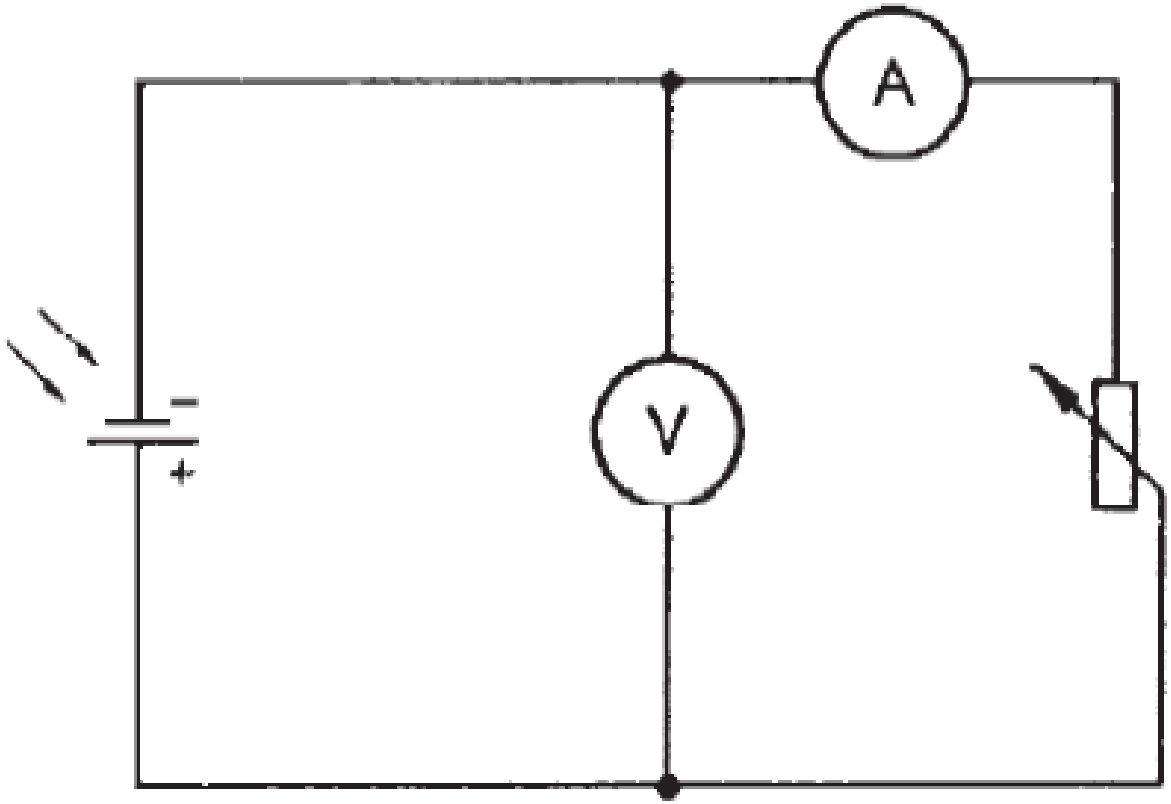
Construction and working:



- Solar cell is the basic unit of solar energy generation system where electrical energy is extracted directly from light energy without any intermediate process.
- The working of a solar cell solely depends upon its photovoltaic effect, hence a solar cell also known as photovoltaic cell.
- A solar cell is basically a semiconductor p-n junction device. It is formed by joining p-type (high concentration of hole or deficiency of electron) and n-type (high concentration of electron) semiconductor material.
- At the junction, excess electrons from n-type try to diffuse to p-side and vice-versa.
- Movement of electrons to the p-side exposes positive ion cores inside, while movement of holes to the n-side exposes negative ion cores in the p-side.
- This results in an electric field at the junction and forming the depletion region.

- When sunlight falls on the solar cell, photons with energy greater than band gap of the semiconductor are absorbed by the cell and generate electron-hole (e-h) pair.
- These e-h pairs migrate respectively to n- and p- side of the pn junction due to electrostatic force of the field across the junction.
- In this way a potential difference is established between two sides of the cell.
- Typically a solar or photovoltaic cell has negative front contact and positive back contact.
- A semiconductor p-n junction is in the middle of these two contacts like a battery.
- If these two sides are connected by an external circuit, current will start flowing from positive to negative terminal of the solar cell. This is basic working principle of a solar cell.
- For silicon, the band gap at room temperature is $E_g = 1.1$ eV and the diffusion potential is $V_D = 0.5$ to 0.7 V.

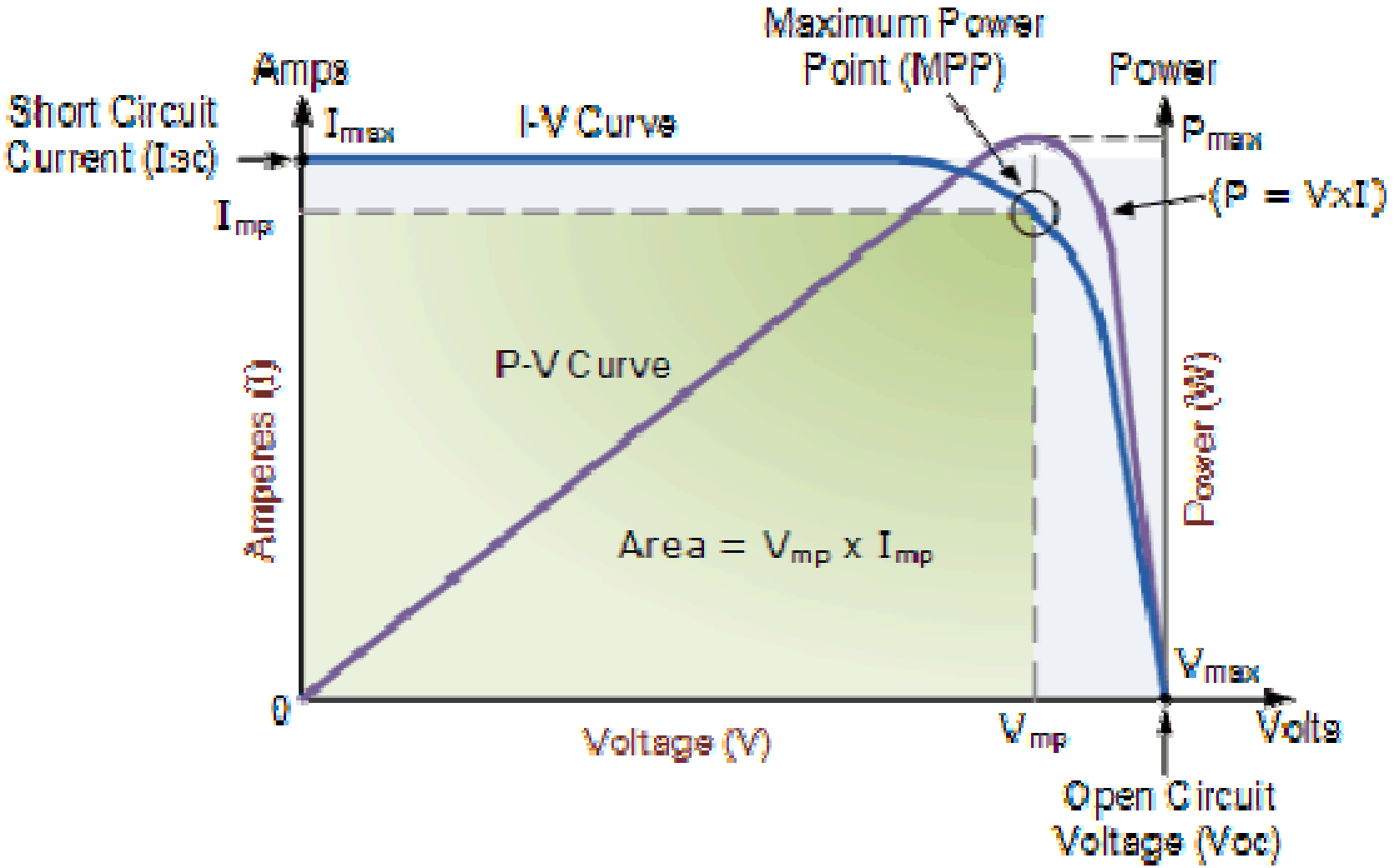
IV Characteristics:



- Solar cell I-V characteristics curve is the superposition of the I-V curves of the solar cell diode in absence (dark) and in presence of light.
- Illuminating a cell adds to the normal dark current in the diode so that the diode law become,

$$I = I_0 \left[\exp\left(\frac{qV}{nkT}\right) - 1 \right] - I_L$$

I_0 = dark saturation current
q = electronic charge
V = applied voltage across the terminals of the diode
n = ideality factor
k = Boltzmann constant
T = temperature
I_L = light generated current


Maximum Power
Point (MPP)
Amps
Power
Short Circuit
Current (Isc)
I_{max}
I-V Curve
P_{max}
$(P = VxI)$
I_{mp}
P-V Curve
Amperes (I)
Power (W)
Area = V_{mp} x I_{mp}
V_{max}
0
Voltage (V)
V_{mp}
Volts
Open Circuit
Voltage (Voc)

- The **short-circuit current** I_{sc} is the current through the solar cell when the voltage across the solar cell is zero (i.e., when the solar cell is short circuited).
- This is due to the generation and collection of light generated carriers.
- For an ideal solar cell at most moderate resistive loss mechanism the short-circuit current and the light- generated current are identical.
- Therefore, the short-circuit current is the largest current which may be drawn from the solar cell.

- The **open circuit voltage**, V_{oc}, is the maximum voltage available from a solar cell, and this occurs at zero current.
- The open – circuit voltage corresponds to the amount of forward bias on the solar cell due to the bias of the solar cell junction with the light-generated current.
- The **fill factor**, more commonly known by its abbreviation FF, is a parameter which, in conjunction with V_{oc} and I_{sc}, determines the maximum power from a solar cell.
- The FF is defined as the ratio of the maximum power from the solar cell to the product of V_{oc} and I_{sc}.
- Graphically, the FF is a measure of the *squareness* of the solar cell and is also the area of the largest rectangle which will fit the IV curve as shown in the figure.

- The **efficiency** is the most commonly used parameter to compare the performance of one solar cell to another.
- Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun.
- In addition to the reflection, the efficiency depends on the spectrum and intensity of the incident sunlight and the temperature of the solar cell.

Example

A circular steel bar of 20 mm diameter carries a tensile load of 30 kN. Find the tensile stress in the bar and the elongation in a length of 300 mm. Given $Y = 2.1 \times 10^5 \text{N/mm}^2$.

$$A = \pi R^2 = 3.142 \times \left(\frac{20 \times 10^{-3}\text{m}}{2}\right)^2 = 0.142 \times 10^{-4}\text{m}^2$$

Tensile stress is,

$$T = \frac{F}{A} = \frac{30 \times 10^3\text{N}}{0.142 \times 10^{-4}\text{m}^2} = 95492965.86\text{Nm}^{-2}$$

Tensile strain have relations,

$$Y = \frac{\text{stress}}{\text{strain}} \qquad \therefore \text{ strain} = \text{stress}/Y$$

and

$$\text{strain} = \frac{\text{increase in length}}{\text{original length}}$$

From these,

$$\text{increase in length} = \text{original length} \times \text{strain} = \text{original length} \times \text{stress}/Y$$

$$\therefore \Delta l = 300 \times 10^{-3}\text{m} \times \frac{95492965.86\text{Nm}^{-2}}{2.1 \times 10^5\text{N}(10^{-3})^{-2}\text{m}^{-2}} = 0.1364 \times 10^{-3}\text{m}$$

Example

The Ultimate shear stress of a mild steel flat is $400\,\mathrm{N/mm^2}$. Calculate the diameter of a hole that can be punched through it without exceeding a compressive stress $200\mathrm{N/mm^2}$ in the punch. Given that the thickness of the flat is $15\mathrm{mm}$. Also, determine the strain in the punch if Young's modulus of its material is $2.1 \times 10^5 \mathrm{N/mm^2}$.

The maximum stress at which a material shears off is known as its ultimate shear stress. Let d be the diameter of the cylindrical hole that will be punched.

Force required to punch the hole = Ultimate shear stress $\times$ Curved surface area of the hole to be sheared

If d is the diameter of the hole and t is the thickness of the flat, the curved surface area of the cylindrical hole will be,

$$A = 2\pi \left(\frac{d}{2}\right) t = \pi t d$$

$$\therefore \; F = 400\,\mathrm{N/mm^2} \times \pi t d$$

Here, the available force = compressive stress applied $\times$ circular face area of the hole

$$\therefore \; F = 200\mathrm{N/mm^2} \times \pi \left(\frac{d}{2}\right)^2$$

Equating these two force expressions,

$$400\,\mathrm{N/mm^2} \times \pi t d = 200\mathrm{N/mm^2} \times \pi \left(\frac{d}{2}\right)^2$$

the diameter of the hole that can be punched with that given force is,

$$d = 8t = 8 \times 15\mathrm{mm} = 120\mathrm{mm}$$

Strain in the punch is,

$$= \frac{\text{stress}}{\text{Young's modulus}} = \frac{200\mathrm{N/mm^2}}{2.1 \times 10^5 \mathrm{N/mm^2}} = 9.5238 \times 10^{-4}$$